Universal gravitational energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the zero-at-infinity convention for gravitational potential energy.

SUCCESS CRITERIA - I can...

1. Use center-to-center distance.
2. Explain why negative potential energy does not mean negative kinetic energy.
3. Compute changes from final minus initial energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Net force rises linearly from zero to 10 N over 0.3 m. Find work and final speed for a 0.3 kg cart starting at rest.

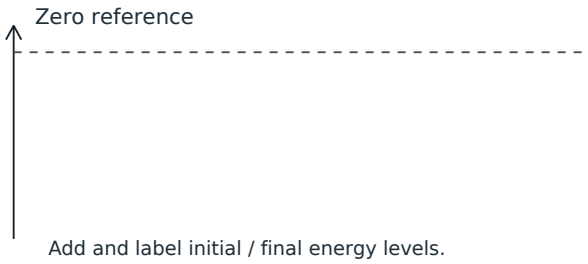
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $U = -GMm/r$; $\Delta U = U_f - U_i$.
Outside a spherical source; choose U = 0 at infinity. Moving outward makes U less negative.

Worked reasoning: For a supplied source with GMm=120 J·m, potential energy is U=-120/r. At r=3 m, U=-40 J; at 6 m, U=-20 J. Moving outward raises U by 20 J.

Your turn: Draw an energy number line and mark the direction toward the zero at infinite separation.



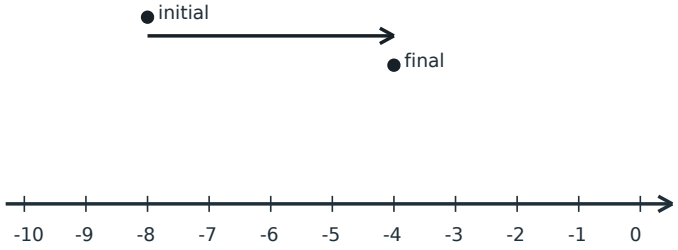
Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: ① ② ③

03 Read and interpret the model

An object moves outward from r to 2 r. Does gravitational potential energy become more or less negative? Draw a number-line comparison.

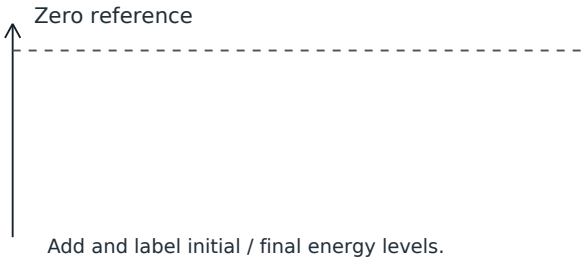


Example: potential energy (J)

Marzano check - circle: ① ② ③

04 Apply the physics

A 1000 kg satellite moves from $r=7.0\times10^6$ to 1.4×10^7 m. Use $GM=4.0\times10^{14}$ m³/s². Find U at each point and ΔU.



Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: ① ② ③

Universal gravitational energy

antocciphysics

Continue down each column.

05 Explain your reasoning

A student says “negative potential energy means the object has negative kinetic energy.” Correct the claim.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

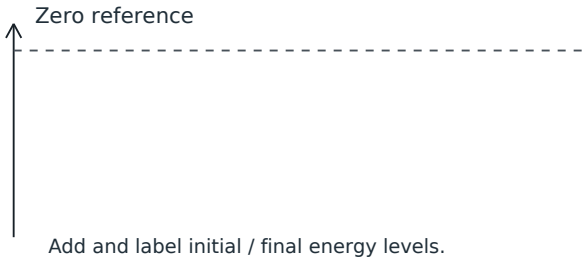
A model target needs a 4 km sideways shift over 6×10^{6} s. Its mass is 8×10^{8} kg. Estimate Δv and impulse; state the model used.
Label the quantities in the model and state its assumptions.

Impulse → Velocity change → Later offset

Marzano check - circle: 1 2 3

07 Exit check

With GMm=60 J·m, move from r=5 to r=15 m. Find work by gravity.



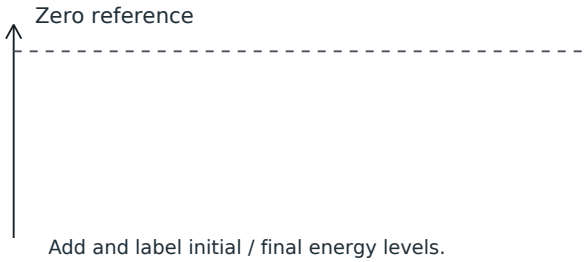
Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A mass moves from r=R to 3 R. Express ΔU and work by gravity in terms of GMm/R. Compare with reaching infinity.



Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: 1 2 3